

# score\_sjsdm\_joint\_tuning\_predictions.R

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| score_sjsdm_joint_tuning_predictions        |  |
| <i>Score Joint sjSDM Tuning Predictions</i> |  |

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## Description

Calculates the held-out joint sjSDM negative log likelihood and probability diagnostics used by regularization tuning.

## Usage

```
score_sjsdm_joint_tuning_predictions(  
  object = NULL,  
  data_test_input = NULL,  
  data_observed = NULL,  
  data_predicted = NULL,  
  epsilon = 1e-06,  
  n_likelihood_draws = 20L,  
  score_seed = 900723L  
)
```

## Arguments

|                               |                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------|
| object                        | Fitted ‘sjSDM’ object containing the model likelihood callback, formulas, and fit settings.            |
| data_test_input               | Fold-local test input containing ‘data_abiotic_to_fit’ and, for spatial models, ‘data_spatial_to_fit’. |
| data_observed, data_predicted | Aligned binary observations and marginal predicted probabilities.                                      |
| epsilon                       | Probability clipping tolerance passed to [score_sjsdm_tuning_predictions()].                           |

`n_likelihood_draws` Positive integer number of stochastic likelihood evaluations to average. Defaults to '20L', matching 'sjSDM::sjSDM\_cv()'.  
`score_seed` Non-negative integer used for deterministic stochastic likelihood draws.

### Details

Test design matrices are reconstructed from the fitted model formulas. Likelihood sampling and batch size are read from the fitted model settings. The caller's R and available PyTorch random-number states are restored.

### Value

One-row tibble with retained taxa, response-value count, total joint negative log likelihood, joint loss per response, and macro AUC.

### Examples

```
## Not run:
score_sjsdm_joint_tuning_predictions(
  object = mod_fit,
  data_test_input = list_test_input,
  data_observed = data_test_observed,
  data_predicted = data_test_predicted
)

## End(Not run)
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